

Major Advanced Topics

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Role of Spin-Orbit Coupling in High-Order Harmonic Generation Revealed by Supercycle Rydberg Trajectories

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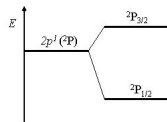
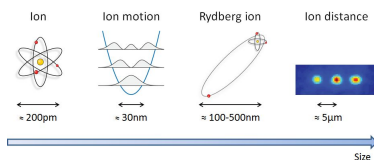
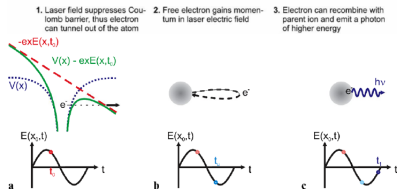
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Role of Spin-Orbit Coupling in High-Order Harmonic Generation Revealed by Supercycle Rydberg Trajectories (Phys. Rev. Lett. 129, 173202 (2022))

Abstract

- High-harmonic generation (HHG) is typically thought as a *sub-laser-cycle* process.
- However, *long-lived Rydberg states* have been suggested to play an important role in HHG generation by the combination of counterrotating circularly polarized fields and its second harmonic.
- This letter is about the experimental evidence of the effect of long and stable Rydberg trajectories in such fields by means of the spin-orbit splitting.
- Observation is at odd with the general view that long-lived Rydberg orbits should generate negligible contribution to the field.



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The spin-orbit coupling

- Particles have two different angular momentum which are the orbital angular momentum L and spin angular momentum S .
- Their coupling known as the spin-orbit coupling is

$$\langle \mathbf{L} \cdot \mathbf{S} \rangle = \frac{\hbar^2}{2} \left(j(j+1) - \ell(\ell+1) - s(s+1) \right), \quad (1)$$

induces the shift

$$\Delta_{SO} = \frac{\beta}{2} \left(j(j+1) - \ell(\ell+1) - s(s+1) \right) \quad (2)$$

onto the energy levels E_n , who otherwise only depend on the principal quantum number n and may be degenerated.

$$E_{n\ell} = E_n \pm \Delta_{SO}, \quad \beta = \beta(n, \ell) = Z^4 \frac{\mu_0}{4\pi} g_s \mu_N^2 \frac{1}{n^3 a_0^3 \ell(\ell+1/2)(\ell+1)}. \quad (3)$$

- Excitation of the spin-orbit split states upon ionization can either induces *ultrafast hole dynamics* or entanglement between the ion and the photoelectron.

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The role of the spin-orbit coupling

- Generally thought as negligible because, for Argon, the delay $\tau = t_r - t_i \sim 1/\omega$ in which you can act upon the hole is too short. (Because $\tau\Delta_{S0} \ll 1$).
- Acting upon the hole means dynamically preventing an early recombination of the electron in the continuum with the ion in order to increase the order of the harmonic.
- In this paper, the authors use the *Larmor clock* to identify the contribution of long-lived Rydberg orbits to the HHG.
- Moreover, they confirm a results saying that these Rydberg states lead to the generation of symmetry forbidden harmonics.

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The point of these authors

- They use Argon for which for which they can affect the precession of the spin of the hole during the period range $T_{SO} = 23.5$ fs.
- Note that T_{SO} is the time necessary for the dynamics of the spin orbit coupling to take effect and not the period of the laser, which is τ .
- If and only if τ spans at least $\tau \sim T_{SO}$, the harmonic lines will be split by Δ_{SO} .
- This is what they observe experimentally with a circularly polarized 800 nm fundamental field and its second harmonic (whose frequency is 2ω).
- They claim that the fact that they can see those lines is due to those supercycle trajectories.

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Long Rydberg trajectories enhance HHG spectra

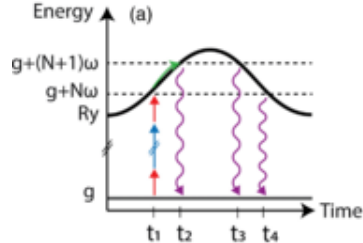
- Figure shows the participation of long-lived Rydberg state to the HHG process.
- They exist because of Freeman resonances which are the frequency-domain counterpart to the time-domain frustrated tunneling.
- Resonance can arise anywhere in the pulse, whenever

$$E_n + U_p - E_g = n_\omega \omega + n_{2\omega} 2\omega. \quad (4)$$

- Resonance emission is liable to constructive cycle-to-cycle interference, when the relative phase

$$\phi = (E_n + U_p - E_g) T_{\text{cyc}} = 2\pi m. \quad (5)$$

- **Forbidden transition harmonic lines are enabled because the ponderomotive energy U_p conserves the parity and the angular momentum.**

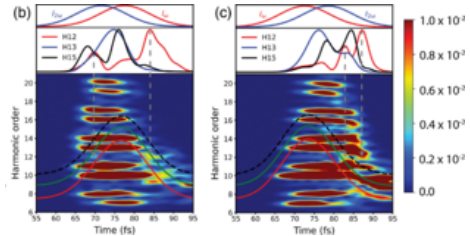


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Long-lived Rydberg orbits allow parity forbidden transitions

- In the case of bicircular $(\omega, 2\omega)$, forbidden harmonics are of order $3N$.
- Figure shows TDSE calculations in Ar, for two gaussian pulses with FWHM of 15 fs and frequency ω and 2ω .

$$\omega = 0.057 \text{ a.u.}, \quad U_p = \sum_{\omega'} U_p^{\omega'} \sim 7\omega.$$

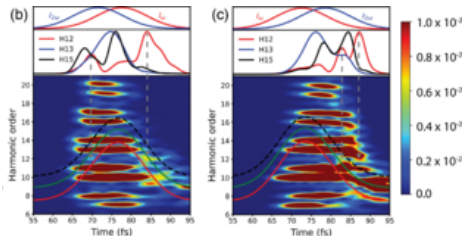


- The Gabor spectrogram shows the two cases where the 2ω laser precedes the ω laser (b) and vice-versa (c).
- Ponderomotive shifts of the ground state, along with the $2p^5 4s$ and $2p^5 3d$ states in black, red and green respectively.
- Top panel represents the integration of the Gabor spectrum in the $\pm 0.25\omega$ range.

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Long-lived Rydberg orbits allow the manipulation of spin-orbit dynamics

- For both H12 and H15, the contribution of the Rydberg state is more visible than for the allowed H13.
- Forbidden lines are made possible by the absorption of counterrotating components from either fields.

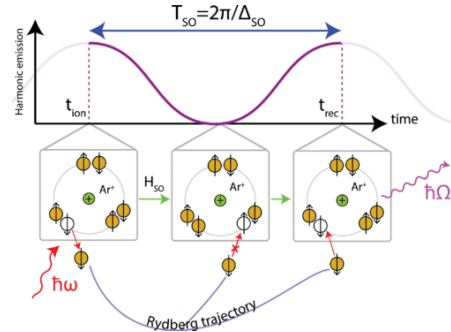


- When 2ω precedes the fundamental pulse, enhancement appears in resonance around the peak of the laser.
- When the fundamental pulse precedes 2ω , enhancement appears around the falling edges.
- The difference between the two cases when the second harmonic precedes or follows the fundamental is called a *temporal gate*.
- Rydberg orbits spending a longer time trapped inside the pulse allow the resolving of the spin orbit hole dynamics.

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The spin-orbit hole dynamics

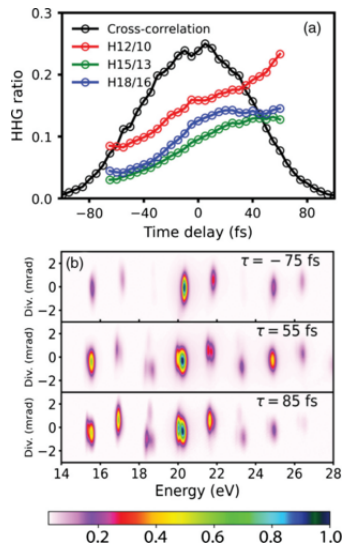
- Spin orbit temporal modulation of the emission leading to spectral splitting of the harmonic lines.
- The photorecombination amplitude is temporally modulated by the factor $1 + \exp(-i\Delta_{SO}\tau)$.
- The point is, they are able to do this experimentally.
- This is a means to an end to enhance and thus control harmonic generation contribution from the Rydberg states.
- Moreover, they use the spin orbit coupling to
 - experimentally control the precession of the hole in the first place,
 - make sure that what they are seeing is indeed the effect of the harmonic.



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Experimental results

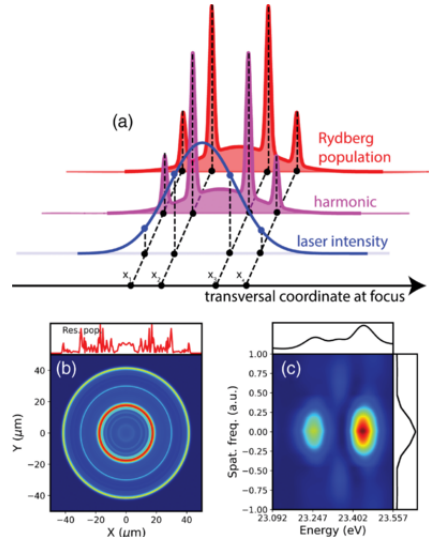
- Figure (a) shows the ration between allowed and forbidden $3N$ following harmonic line for a 2ω preceding field.
- The delay dependence of the ratio confirms that symmetry forbidden harmonic lines appearing on the falling edge.
- Figure (b) shows "the far field spatially resolved" harmonic spectrum at positive and negative delays.
- At positive delays, strong enhancement of forbidden lines can be seen.
- Furthermore, the spectral splitting of most lines is visible.
- Absence of the spectral lines for negative time delay is consistent with discussion.



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Experimental results

- Figure (a) shows role of the spatiotemporal gating in the generation of harmonics.
- The Resonance with Stark-shifted Rydberg energies lead to resonant enhancement for a given harmonic at x_i .
- Figure (b) shows "the near-field" profile of H15.
- Upper panel shows the spatial distribution of the residual population in the excited state. Notice that profile looks like panel (a).
- Figure (c) shows the spin-orbit splitting of H15 in the far-field.
- Top panel shows frequency-resolved spatially-integrated distribution when spin-orbit modulations are included.
- Right panel shows frequency-integrated spatially-resolved H15 line.



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Conclusion

- This letter is about the effect of long Rydberg trajectories onto the HHG spectra for the Ar.
 - The harmonic lines are tracked using the spin orbit splitting and the presence of forbidden lines in the case of counterrotating fundamental with its second harmonic.
 - This is made possible by the fact that hole dynamics can be acted upon in the case of spin-orbit splitting by unclear means.
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- The letter sheds new light on the role of excited states in general in the HHG process.
 - Since the timescale can be acted upon through spin-orbit coupling, experimentalists get to choose between
 - delayed emission of the pulse train,
 - delayed hyper Raman lines in some conditions,
 - coherent XUV free-induction decay.
 - This is an important step for extending HHG spectroscopy beyond . the optical cycle